

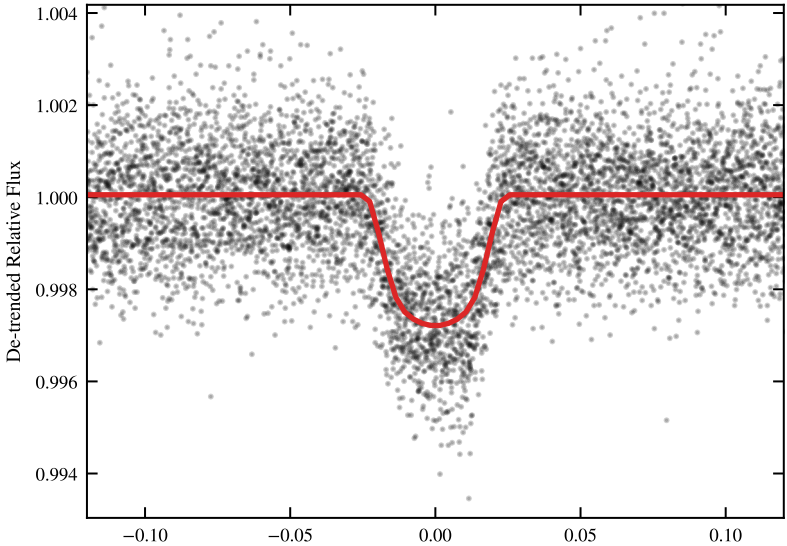
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 148914726 | RA: 187.05590° | Dec: 67.24004° | Sectors: 14, 15

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	1.2615 ± 1.72e-05
Rp/R★	0.05964 ± 0.00277
Rp (R⊕)	5.0562 ± 0.337
b	0.88823 ± 0.0281
T14 (h)	1.1521 ± 0.0272
a (AU)	0.017844 ± 0.00151
Teq (K)	1570.5 ± 62.7

Primary Phased Transit



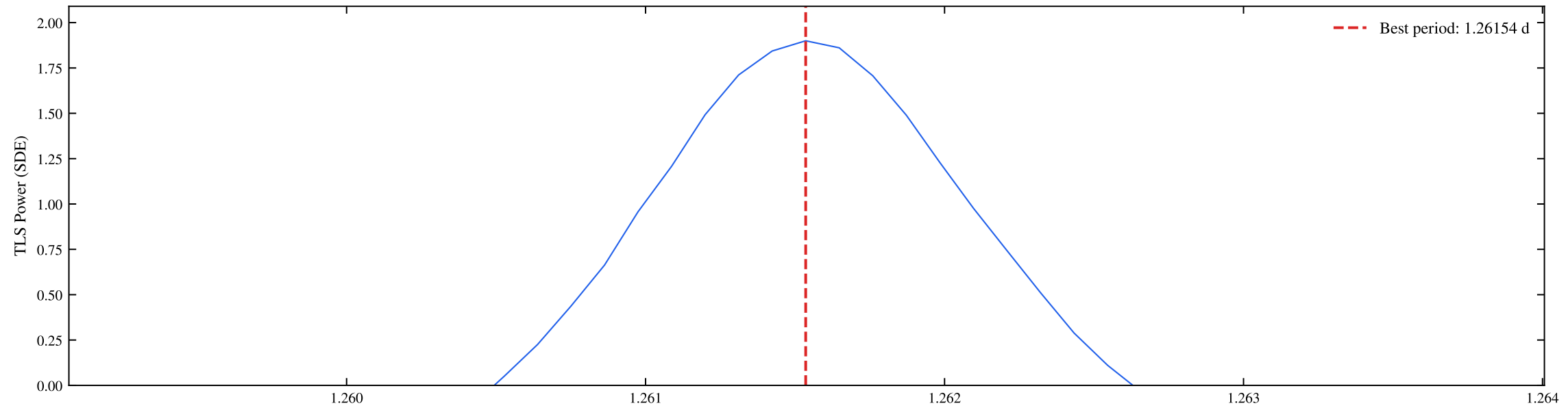
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	1.26154	1.261541 ± 0.00002
P1 Epoch (BTJD)	N/A	1683.4141 ± 0.0004
Max R-hat (R)	N/A	1.0009
Min ESS	N/A	5233
MCMC Divergences	N/A	0

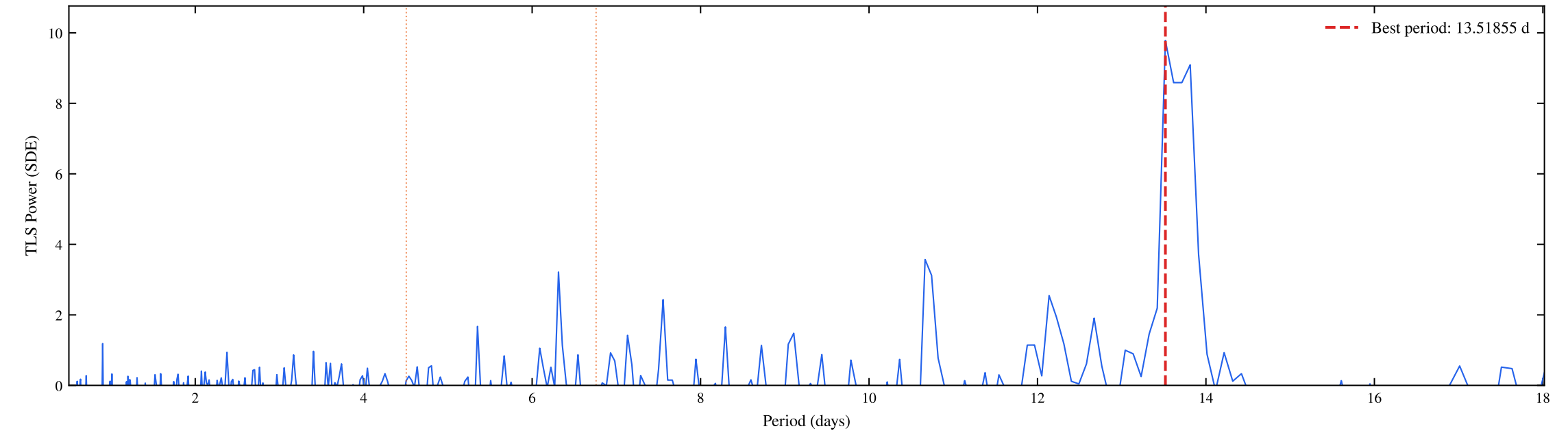
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	0.775 ± 0.037	VizieR TIC8
M★ (M☉)	0.81 ± 0.08	VizieR TIC8
Teff (K)	4944 ± 103	VizieR TIC8
log g (dex)	4.568 ± 0.112	VizieR TIC8
[Fe/H]	0.0997 ± 0.00825	Gaia DR3
ρ★ (g/cm³)	2.453 ± 0.427	Derived

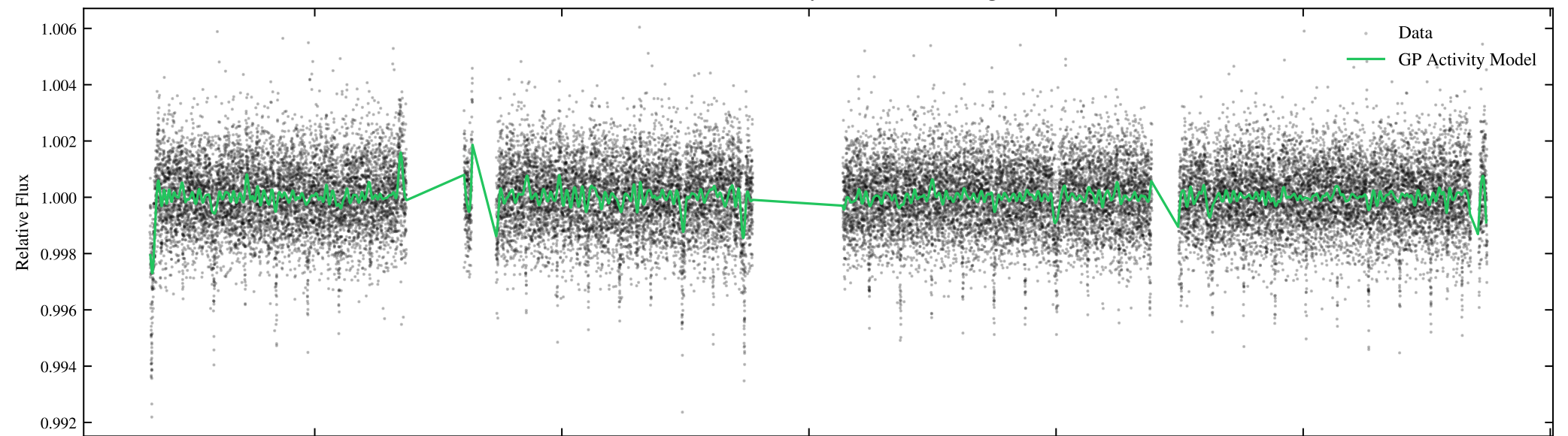
TIC 148914726 | Sectors: 14, 15 | TLS Periodogram (Planet 1 Search) (SDE = 1.90)



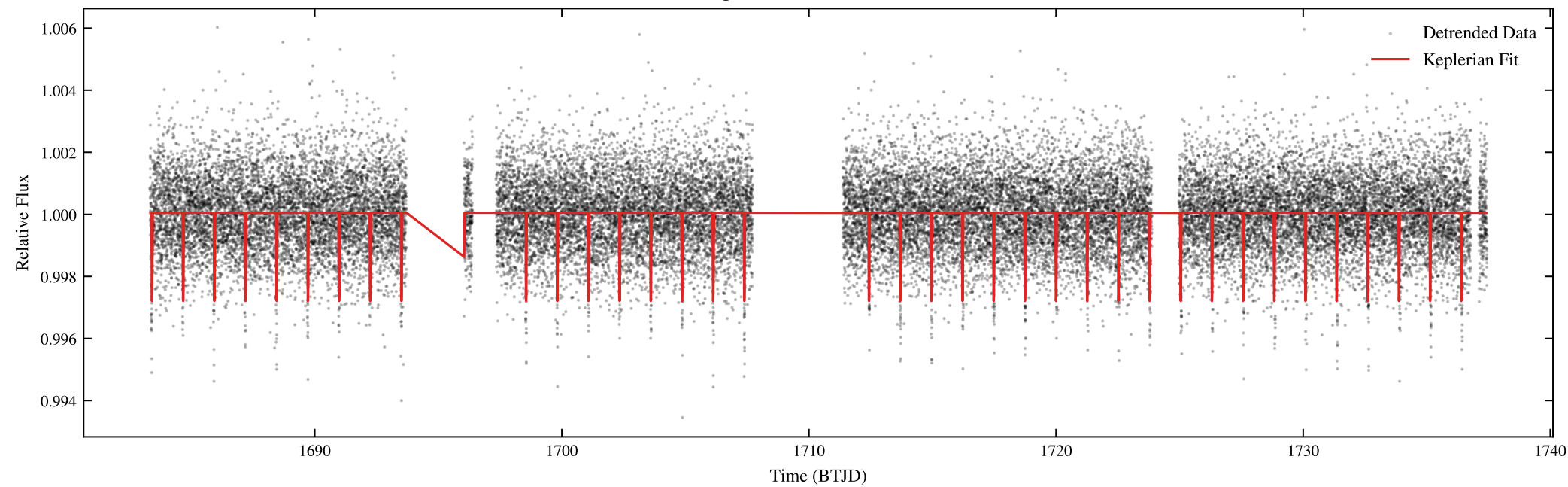
TIC 148914726 | Sectors: 14, 15 | TLS Periodogram (Planet 2 Search) (SDE = 9.78)



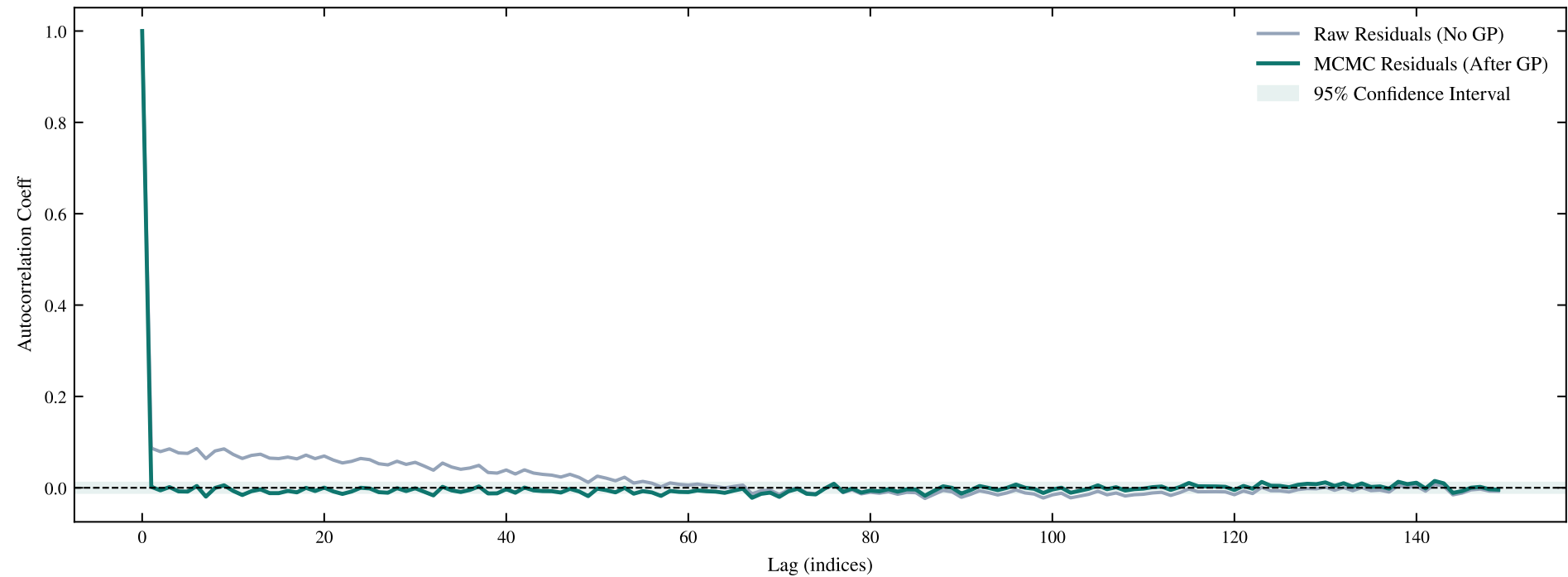
TIC 148914726 | Sectors: 14, 15 | Bayesian GP Detrending (Time Domain)



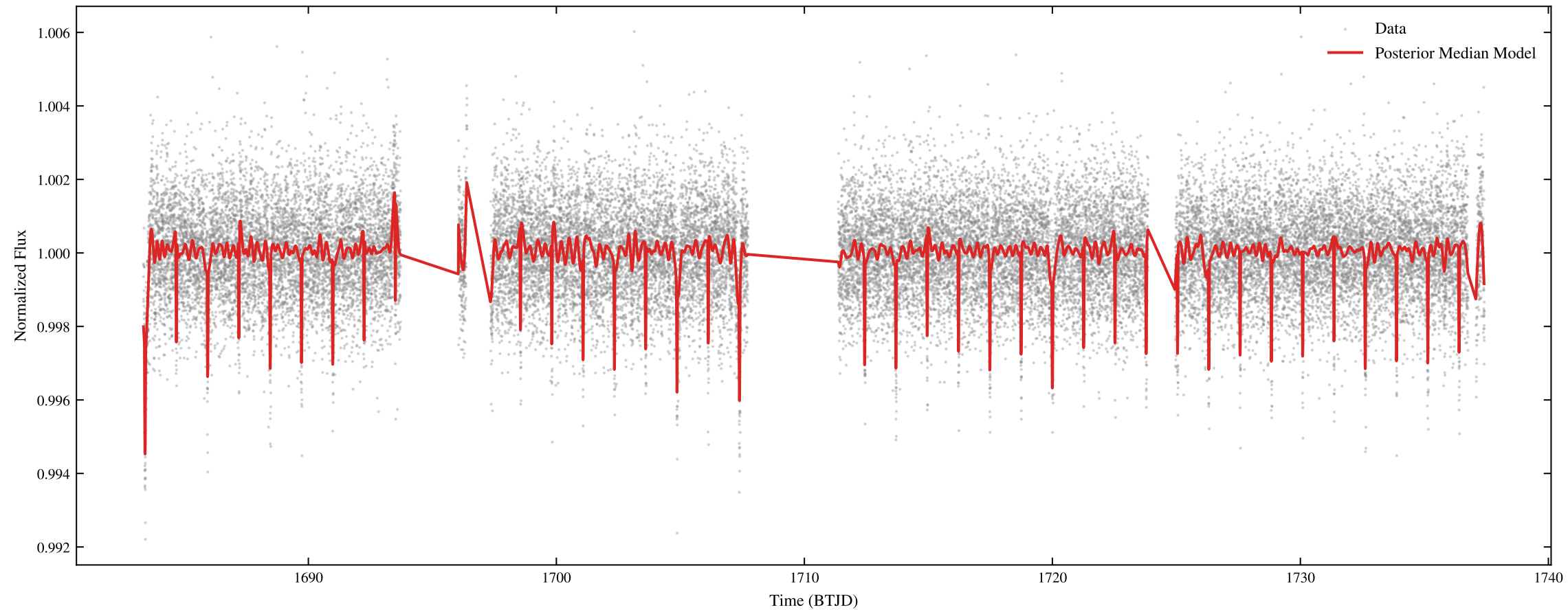
Detrended Light Curve & Best-Fit Transit Model

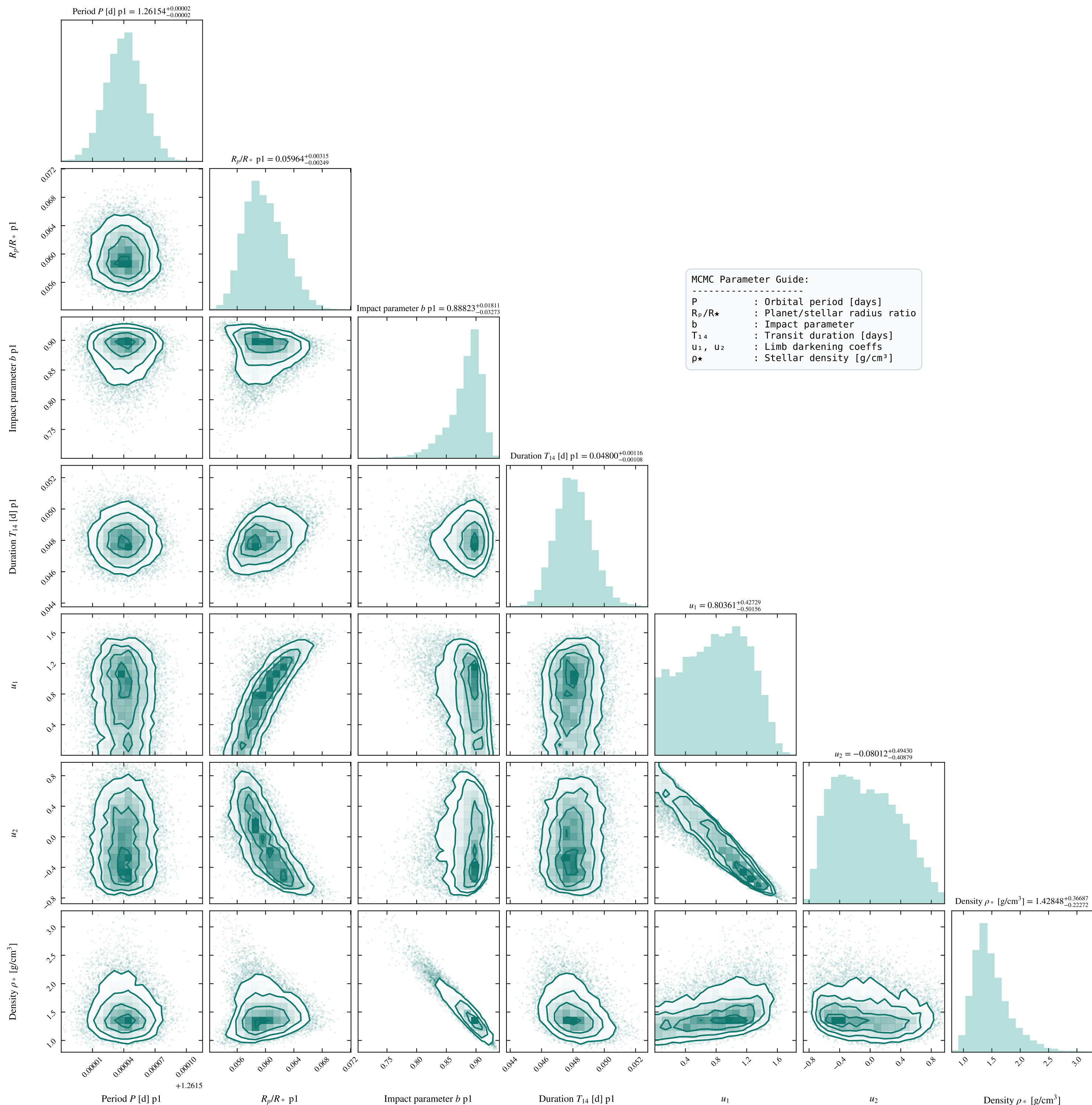


TIC 148914726 | Sectors: 14, 15 | GP Residuals Autocorrelation Function (ACF)

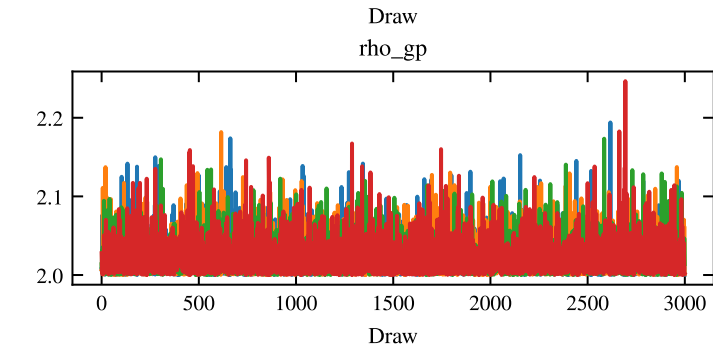
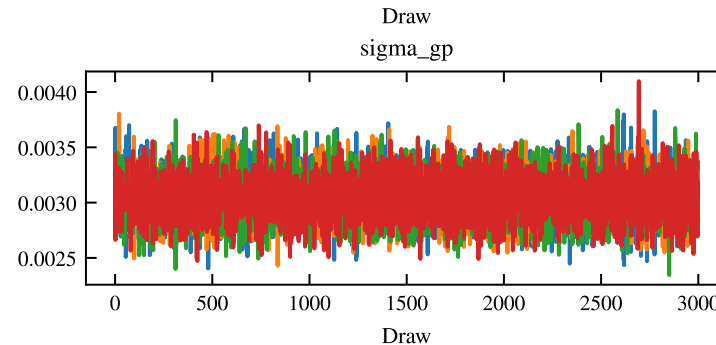
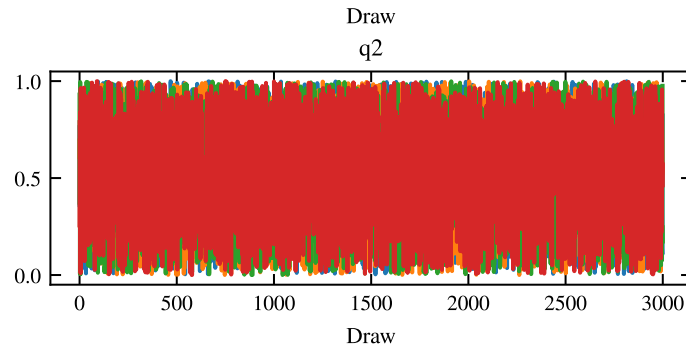
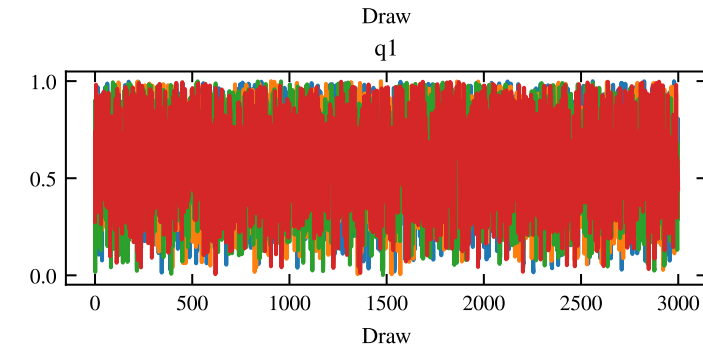
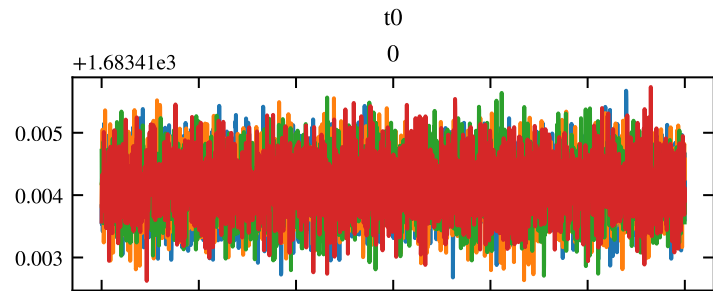
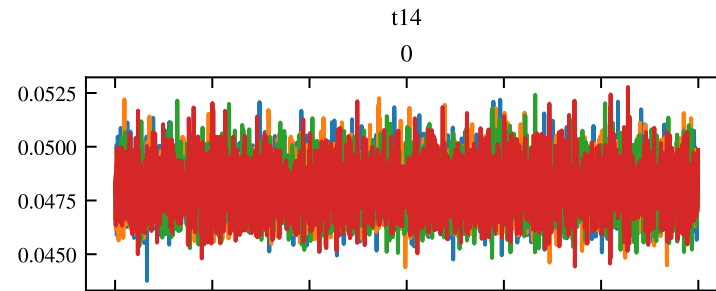
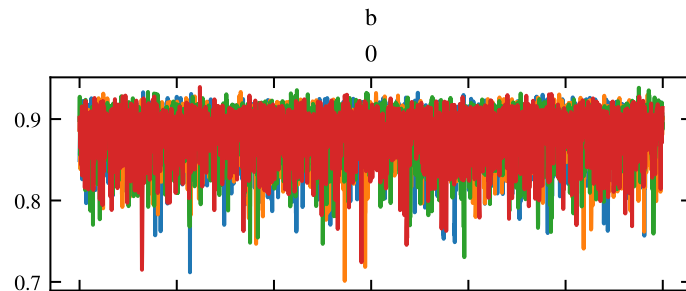
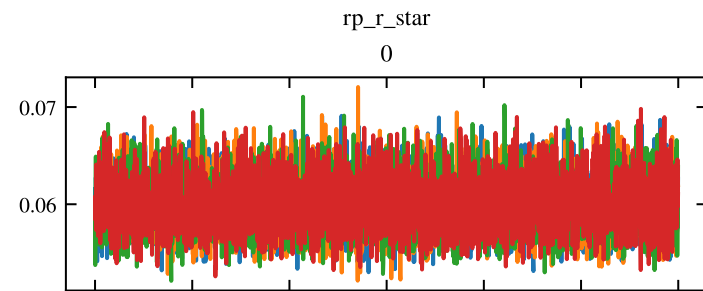


TIC 148914726 | Sectors: 14, 15 | MCMC Posterior Predictive Check

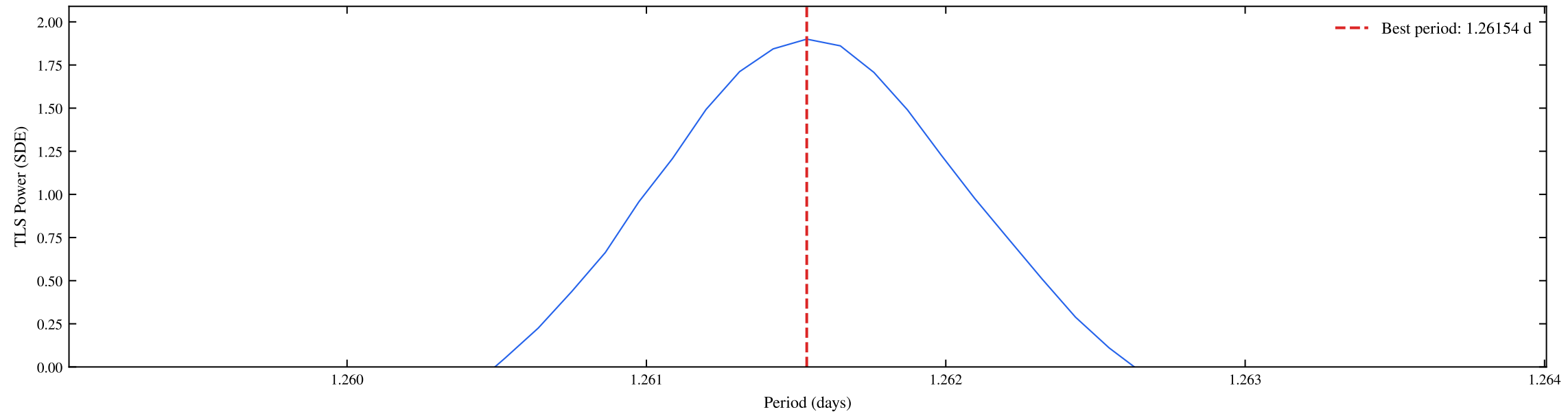




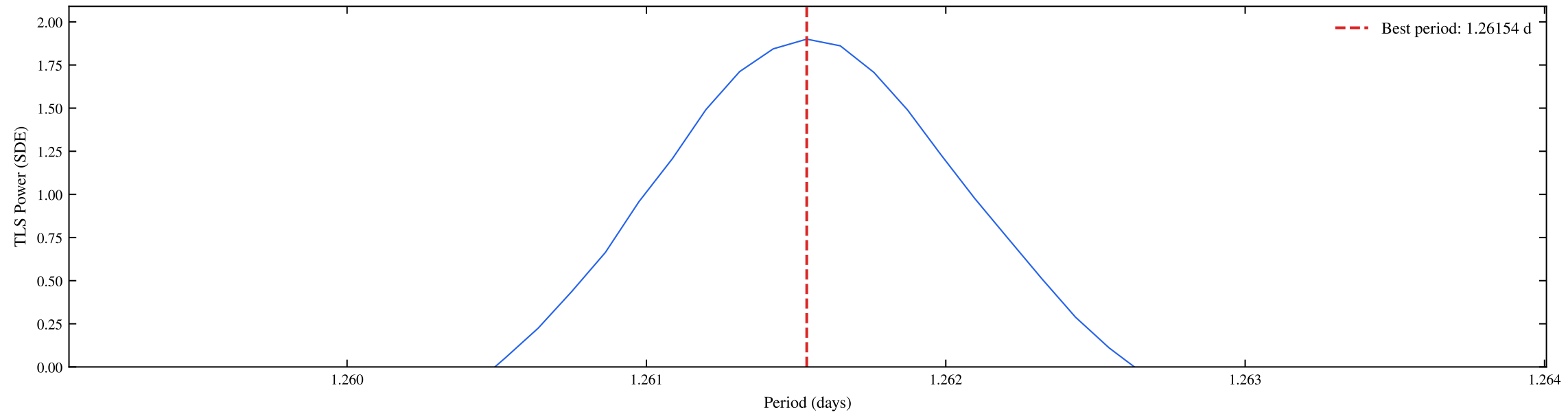
TIC 148914726 | Sectors: 14, 15 | MCMC Trace Plots



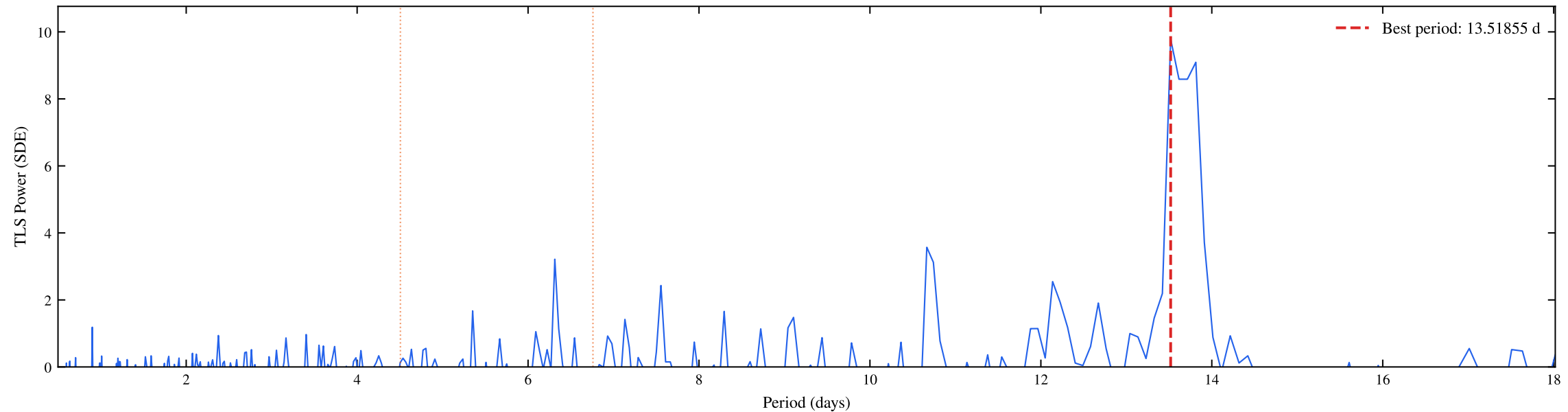
TIC 148914726 | Sectors: 14, 15 | TLS Coarse Periodogram (SDE = 1.90)



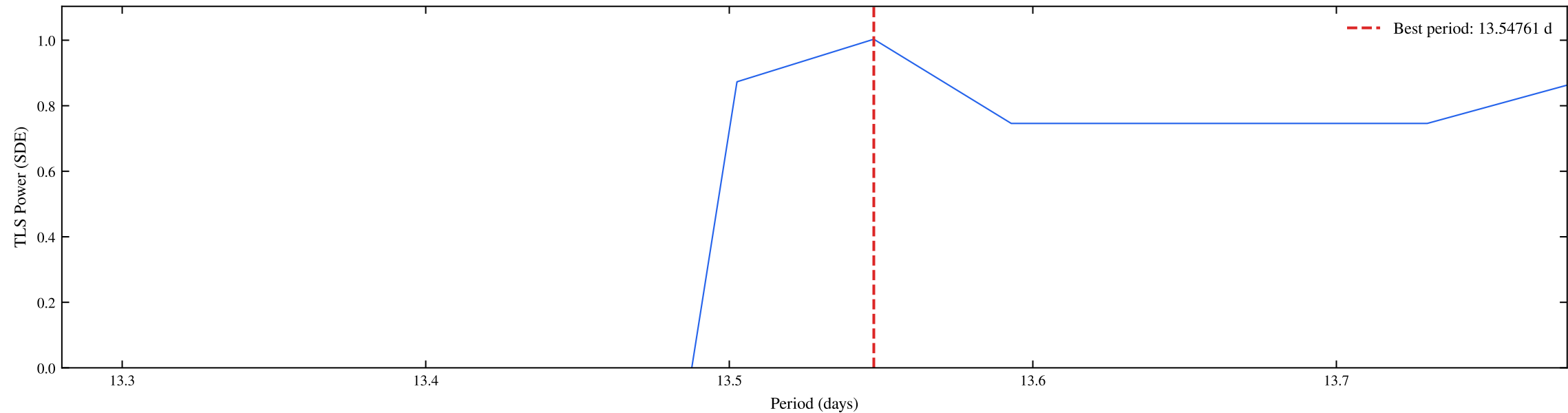
TIC 148914726 | Sectors: 14, 15 | TLS Fine Periodogram (SDE = 1.90)



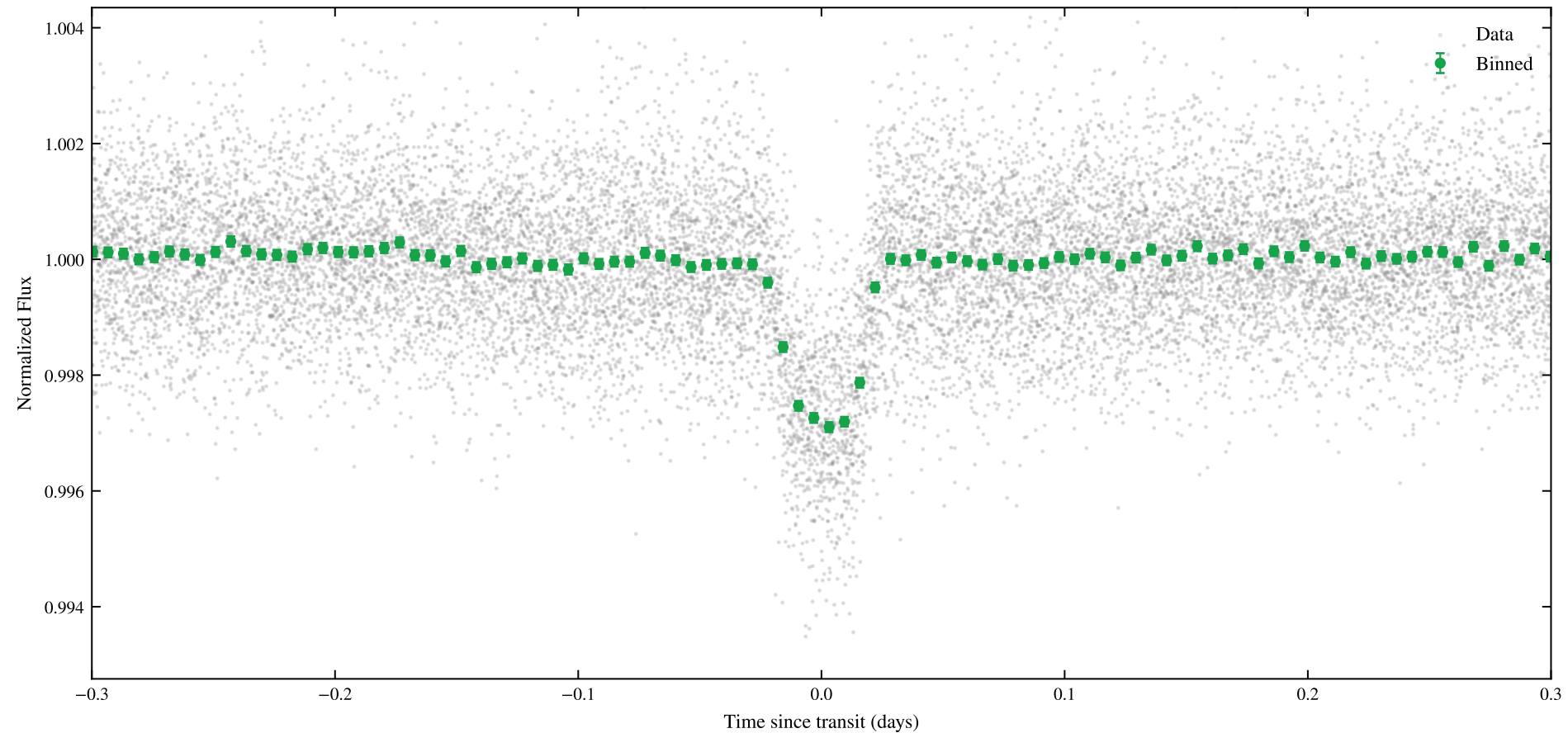
TIC 148914726 | Sectors: 14, 15 | TLS Coarse Periodogram (SDE = 9.78)



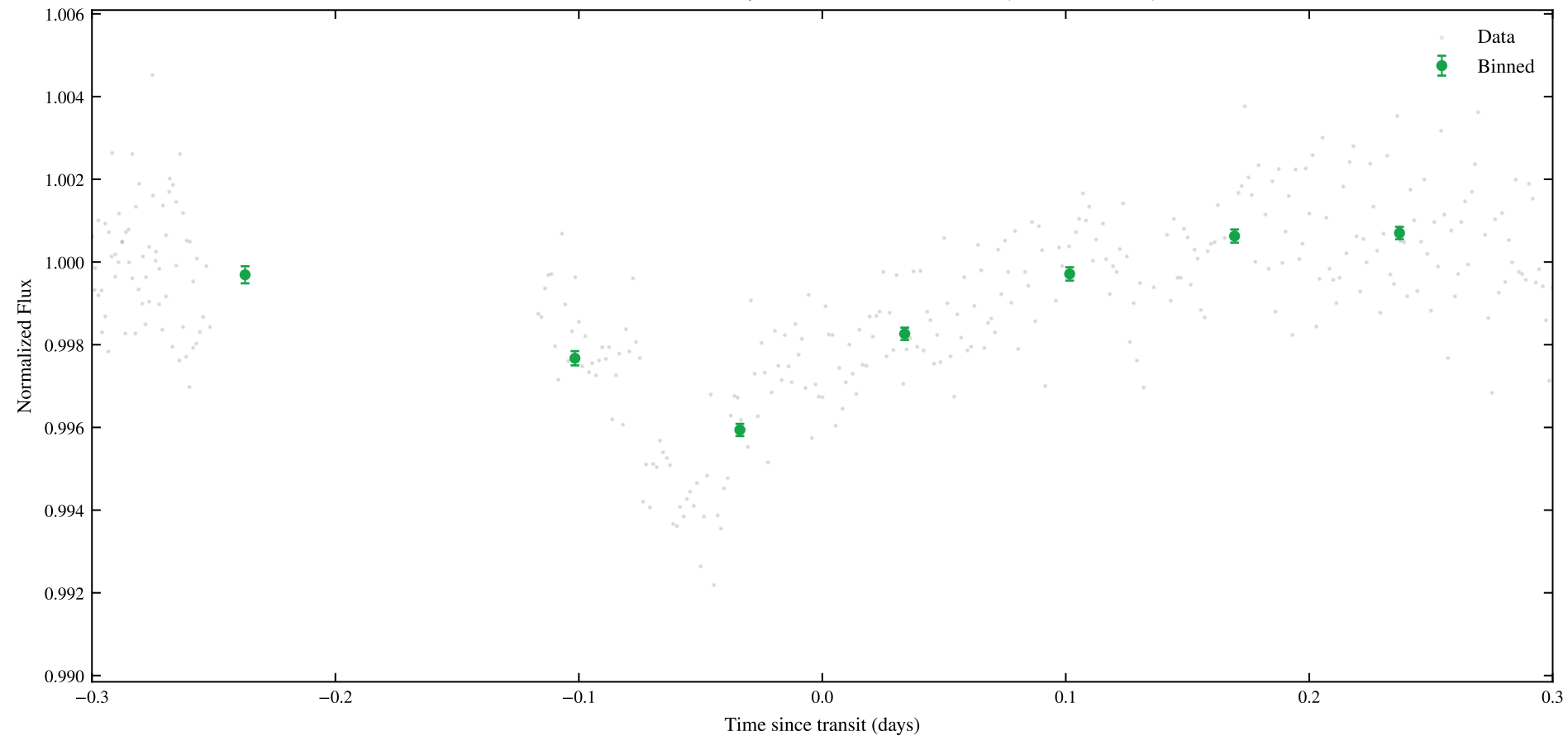
TIC 148914726 | Sectors: 14, 15 | TLS Fine Periodogram (SDE = 1.00)



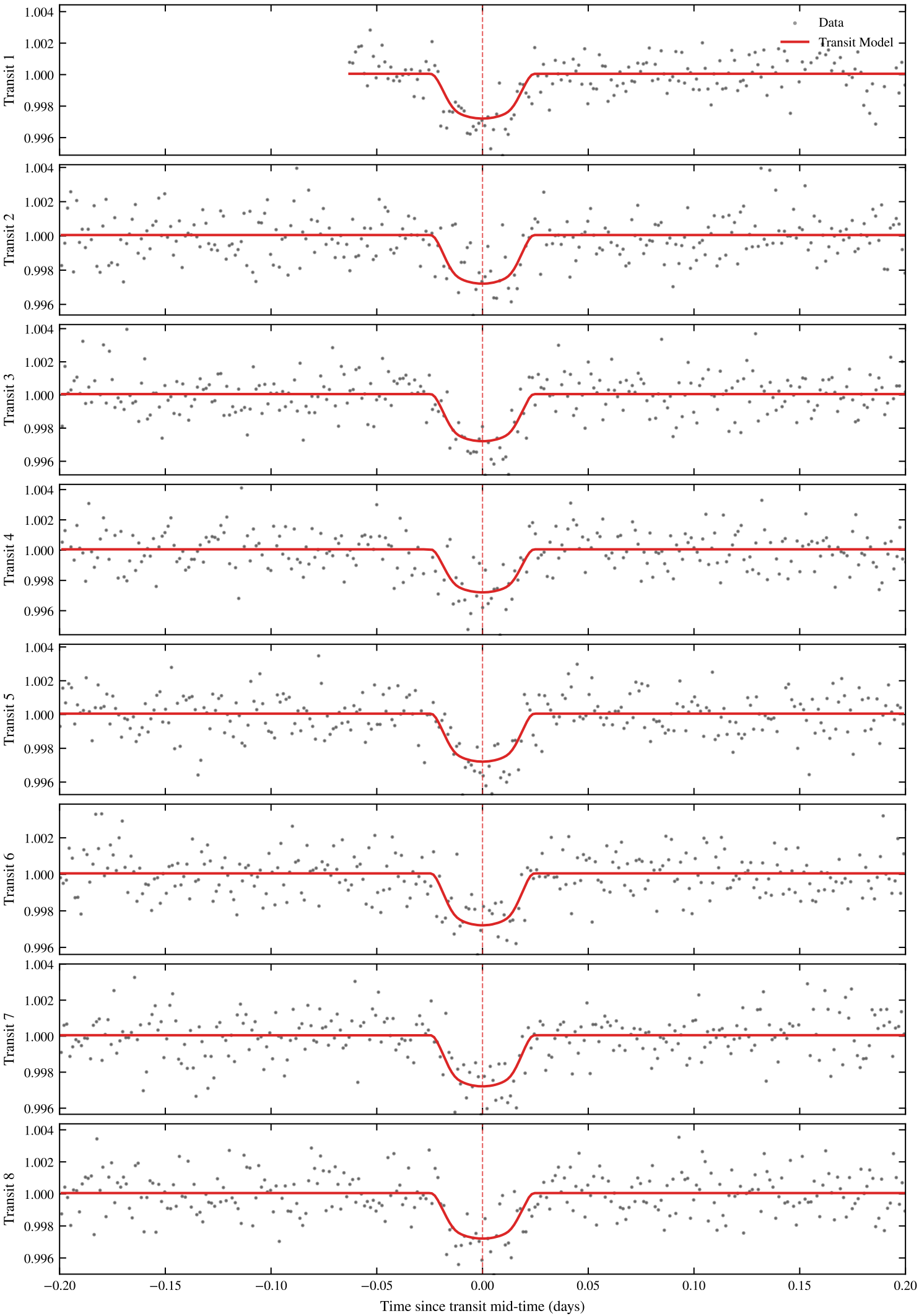
TIC 148914726 | Sectors: 14, 15 | Phase-Folded Transit (P = 1.26154 d)



TIC 148914726 | Sectors: 14, 15 | Phase-Folded Transit (P = 13.54761 d)



TIC 148914726 | Sectors: 14, 15 | Stacked Individual Transits



TIC 148914726 | Sectors: 14, 15 | Phased MCMC Fit - Planet 1

